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Project Title:	Gravitational Influence on Cerebellar Control of Gaze Movement and Adaptation		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
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Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 5/2009) NOTE: Received NCE to 1/31/2008 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Fuchs, Albert (University of Washington) Robinson, Farrel (University of Washington)		
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Task Description:	Microgravity alters the normal relationships between coordinated multi-axial body movements and sensory feedback. As subjects undergo transitions between 1-G to near 0- G, they experience inappropriate eye and head movement and disorientation when shifting gaze. These changes pose a substantial challenge to mission objectives. One effort to address such challenges has been to develop context dependent pre-adaptation training protocols to train astronauts to quickly adapt based on inertial cues. Unfortunately, while this approach holds much promise, little is known about the reference frame for such adaptation or the underlying physiological mechanisms responsible for these adaptive changes. This black box approach limits progress toward the development of appropriate countermeasures. The aim of this investigation was to develop a laboratory analog of the altered gravitational/inertial environments, and then use identical psychophysical tests to evaluate behavioral performance of human and animal subjects in these environments to define both the neural mechanism and the crucial factors determining adaptation. This research consisted of three components and several sub-projects. First, we described the characteristics of context dependent human adaptation and its transfer to other forms of adaptation, seeking to localize behaviorally the site of adaptation in the CNS. Second, we performed identical experiments on humans and on rhesus monkeys while recording the neural activity of cells within the caudal fastigial nucleus and target brainstem nuclei, which are thought to be important for saccadic adaptation. Finally, we simulated changes in the vestibular environment by changing the activity of the vestibular end organ in monkeys to study context dependent adaptation and the neural discharge associated with these changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has a significant impact on life on earth because it contributes to our knowledge of how the brain controls movement and adapts those movements to challenges faced by novel environments, injury and disease. Eye movements are critically important to everyday human activity because the eye must continually scan the visual world for us to see the world clearly. These eye movements must be fast and accurate, and the orientation of the eye must be maintained in the face of self motion and motion of objects in the visual environment. Similarly, the vestibular sensory system performs a critical function in allowing us to remain stable and oriented in the world. These systems are constantly adapted to allow them to perform their critical roles, and this research addresses the nature of the neural mechanisms that underlie that adaptation. In addition, adaptation procedures developed in this grant may be useful in promoting recovery of behavioral and sensory function following injury. Also, technology derived from this research, including new clinical eye measurement technology and more recently approaches to develop and evaluate a vestibular neural prosthesis will benefit patients with oculomotor and vestibular disorders.
	Conclusions for Human Experiments Human Experiment 1 - (Fuchs, Ling and Phillips). Characterization of saccades made in pitch inverted and roll orientations. Conclusions: These results imply that human eye movements in both the horizontal and vertical plane are affected by changes in whole body orientation. Human Experiment 2 - (Fuchs and Hopp). Identifying sites of saccade amplitude plasticity in humans: transfer of adaptation between different types of saccade. Conclusions: These results, combined with other data, suggest that there are at least two sites of adaptation within the saccadic system. One is used commonly in the generation of express, targeting, delayed and memory-guided saccades, whereas the other is specific for the generation of memory-guided saccades. The relevance of this work is that it informs operational implementation of a preflight adaptation countermeasure. These provide both a benefit and a challenge for operational implementation of a preflight adaptation countermeasure. We have evidence that there is transfer between adaptation of specific saccade types and other saccade types that were not specifically adapted in the laboratory but which would presumably be utilized during spaceflight. We also now know, however, that human saccade adaptation does not transfer as widely across saccade types as does non-human primate saccade adaptation. Human Experiment 3 - (Fuchs and Hopp). Amplitude adaptation occurs where a saccade is represented as a vector and not as its components. Conclusions: This work defines the location of the site of adaptation. This data is consistent with a site of adaptation that is upstream of the brainstem burst generator. Also, adaptation of human saccades with a given horizontal or vertical component will not transfer to all saccades with components of comparable amplitude, but rather only to saccades with a comparable vector. Therefore, multiple preflight adaptations are likely to be required to serve as an effective countermeasure for saccades of different types, amplitudes and directions. While the precise specificity remains to be determined, we now know that component adaptation will not be an effective shortcut to the development of generalized adaptation states. Human Experiment 4- (Fuchs, Ling, Hopp, and Phillips). Context Dependence of Human Saccade Adaptation: Head free versus head fixed gaze shifts. Conclusions: This result has important operational significance because it suggests that a human preflight saccade adaptation countermeasure will transfer to the context in which natural head unrestrained gaze movements are being performed. Such transfer is critical to any useful countermeasure, since astronauts will not be performing saccadic eye movements under laboratory conditions during 1G to 0G transitions, but rather will be performing natural head unrestrained gaze shifts. Human Experiment 5- (Phillips, Ling, Kampel). Kinematics of head upright versus roll tilted head unrestrained gaze shifts. Conclusions: These results suggest that we see and react to the world differently when our trunks are tilted compared with an ordinary head upright condition. We suggest that otolith input affects gaze shifts, which is consistent with observed changes in gaze shifts during transition from 1G to 0G.

Task Progress:

Conclusions for Animal Experiments

Animal Experiment 1. (Phillips, Fuchs, Ling, Newlands). Gaze movement behavior following elimination of semicircular canal input.

Conclusions: This data suggests that vestibular signals are important for the normal control of gaze movement, but that new strategies emerge, involving adaptation of saccade amplitude, to compensate for the loss of vestibular input and maintain accurate gaze movements after canal plugging. This suggests further that the gaze movement system is highly plastic and can adapt to the loss of a critical vestibular input. These plastic changes produce movements that are not normal, but accomplish the task of orienting the eye to objects of interest in the visual environment.

Animal experiment 2 (Phillips, Ling, Fuchs). Behavior of brainstem neurons during head unrestrained gaze shifts following canal plugging.

Conclusions: An efference copy of the head movement may emerge in the discharge of brainstem burst neurons, but not in vestibular nucleus neurons, to produce the observed eye counter-rotations during gaze shifts after loss of vestibular input. The traditional elements that drive and adjust gaze shifts lose control of the end of the gaze shift following vestibular loss. These data are critical for assessing the importance of vestibular signals during gaze shifts with normal and abnormal vestibular contributions. In addition, they suggest that the signature of brainstem saccadic eye movement related neural activity is dramatically altered by loss of vestibular input, leading to the emergence of new mechanisms for accurate gaze control.

Animal Experiment 3. (Fuchs, Ling, and Phillips). CFN activity during head unrestrained gaze shifts in the rhesus monkey.

Conclusions: Neurons in the CFN could carry the vestibular signals requisite for the control of coordinated gaze movement during head unrestrained gaze shifts. Likewise, these cells are possible targets for the vestibular inputs that may be required to establish the context for context dependent adaptation.

Animal experiment 4. (Fuchs, Ling and Phillips). Inactivation of CFN neurons during head unrestrained gaze shifts and head restrained saccades.

Conclusions: The CFN appears to act on the eye movement component of the head free gaze shifts, producing normal velocity and duration characteristics and accurate movements. Inactivation of the CFN produces changes in the eye and gaze components of the gaze shift. These results are consistent with previous models of the CFN as an adaptive controller for gaze movement. This data is not consistent with a role for the CFN in the control of head movement per se.

Animal Experiment 5. (Noto, Robinson and Phillips). Effects of otolith inputs on saccadic eye movement behavior in monkeys.

Conclusions: This data suggests that vertical skew deviation does accompany roll tilt of approximately 45 degrees in monkeys during viewing of near point visual stimuli. This result suggests that a vertical diplopia exists during roll tilt in rhesus monkeys which is uncompensated by normal short-term vestibulo-ocular mechanisms. This could pose a significant challenge to mission objectives if similar dissociations are present in humans under similar conditions, and if the diplopia compromises vision under some visual stimulus conditions, i.e., reduced lighting or required fixation of weakly illuminated avionics during disorienting 1G to 0G transitions.

Animal experiment 6. (Phillips, Noto and Robinson). Saccade adaptation is organized in craniotopic coordinates.

Conclusions: The limited adaptation field with respect to the head suggests an important limitation to preflight adaptation paradigms as a countermeasure. Since the adaptation is restricted to specific vectors with respect to the head, the range of vector directions must be extended during preflight training to allow for adequate representation of visual space during natural movements.

Animal experiment 7 (Phillips, Noto and Robinson). Transfer of saccade adaptation to different tilt orientations.

Conclusions: This result suggests that adaptation training as a countermeasure will transfer to a range of tilt orientations or inertial contexts.

Animal Experiment 8. (Fuchs, Ling, and Phillips). Activity of the basal interstitial nucleus of the cerebellum during head unrestrained gaze shifts.

Conclusions: This result identifies a potential site for the neural control of eye, head, and hand coordination. If this region is involved with coordinated adaptive control of these motor systems, it would represent an important target for pre-flight counter measures to facilitate adaptation of critical coordinated motor task performance during spaceflight (such as coordination of eye, head and hand movements during launch and reentry maneuvers).

Animal Experiment 9- (Phillips, Noto, Ibarreta, Robinson, and Andrews). Context dependent adaptation in the monkey.

Conclusions: Since roll tilt produces vertical skew deviation in the two eyes and pitch cues do not, it is possible that oculomotor torsion or diplopia provide the context for context dependent adaptation and not the vestibular input per se. This result may have profound implications for the application of context dependent saccade adaptation paradigms as a countermeasure, since the implication is that the primary context cue is oculomotor or visual, not vestibular.

Related Research Activity

Related activity. (Kaneko, Phillips) Vestibular and optokinetic testing for research and clinic.

In year one we learned that the major supplier of scleral search coils for human eye movement recording was suspending production of the devices. We responded by purchasing sufficient supplies of the devices to perform our experiments for the full 4 years of our funding. In addition, we developed, with a local firm, a printed circuit device that could serve as a cheap alternative to the existing eye movement transducers.

Conclusions: We successfully developed a prototype eye coil with the assistance of a local biotechnology company, MicroConnex. The ongoing development of this technology is currently supported by NIH/NIDCD.

Dr. Phillips and Dr. Kaneko are working with MicroConnex Corp. on an SBIR application for development of a clinical

	grade version of this device, which is scaled up from the smaller coil pictured above. This funding and continued development of this technology is a direct outgrowth of our experimental research funded through this project. Related activity. (Phillips, Reschke, Harm, Paloski, Krnavek, Somers, Ford). The Effect of Microgravity on Horizontal Saccade Dynamics. Our ground based research compliments flight research begun many years ago at Johnson Space Center by Reschke and colleagues. Dr. Reschke recorded pre-flight, flight and post-flight saccadic eye movement data in astronauts during several space missions. We reexamined much of this very important flight data to better understand the nature of gaze and eye movement changes related to adaptation to novel inertial environments. Conclusions: The flight data confirms that quantitative changes do take place in the metrics of even head restrained gaze shifts during the transition to and from 1 G to near 0 G. It is critical that we understand nature of the changes in gaze movement control during spaceflight to optimize strategies for applying any proposed countermeasures. Related Activity: (Phillips, Fuchs, Ling). Neurophysiological studies of electrical stimulation for the vestibular nerve. An important outgrowth of our research activities and a new research direction for our laboratory was the submission and subsequent funding of an NIH / NIDCD contract proposal to develop a vestibular prosthesis based on existing cochlear implant technology. Dr. Phillips is the P.I. on the contract, which utilizes the approaches described and utilized in this project, including canal plugging and gentamicin injection, and neural and behavioral recording of adaptive recovery of vestibular function during natural movements in rhesus monkeys. This proposal would not have been possible without the knowledge and preliminary results acquired through this NASA funded research. The development of a neural prosthesis has potentially significant implications for the neurovestibular countermeasures program at NASA, because this technology could ultimately be used to tune vestibular responses to the unique inertial and gravitation environments encountered in 0G spaceflight. In addition, the contract will support studies that will further our understanding of the neural mechanisms underlying vestibular adaptation, and in that regard, are a continuation of this NASA funded activity. [NOTE: Above added by Task Book editor from final report; 6/2009]
Bibliography Type:	Description: (Last Updated: 05/13/2009)
Abstracts for Journals and Proceedings	Ling L, Fuchs AF, Ibarreta M, Phillips JO. "Activity of the basal interstitial nucleus of the cerebellum during head unrestrained gaze shifts." Neuroscience 2007, Society for Neuroscience 37th Annual Meeting, San Diego, CA, November 2007. Neuroscience 2007, Society for Neuroscience 37th Annual Meeting, San Diego, CA, November 2007. Abstract 178.11. , Nov-2007